Model Summaries

Name	Summary	Reference
RNN	A recurrent neural network classifier using an RNN backbone for sequential data processing. It can handle sequential inputs and is efficient for tasks like time-series and text data classification.	[SSB+17]
GRU	A classifier using a Gated Recurrent Unit (GRU) backbone. GRUs are a variant of RNNs that can capture long-term dependencies in sequential data, offering better performance than vanilla RNNs.	[CGCB14]
MLP	A multi-layer perceptron (MLP) classifier with a fully connected network backbone. It is commonly used for tasks where the data does not have a sequential or temporal structure.	[PBPPM09]
LSTM	A classifier using a Long Short-Term Memory (LSTM) backbone. LSTMs are a type of RNN that can capture long-term dependencies and handle vanishing gradient problems in deep sequences.	[Gra12]
FNet	Uses an FNet backbone for processing input data. FNet employs a Fourier transformation in place of attention mechanisms, offering a more efficient way to model relationships between inputs.	[LTAEO21]
gMLP	A classifier using the gMLP (Gated MLP) backbone, which is a type of multi-layer perceptron designed to model long-range dependencies using gated transformations.	[LDSL21]
Mixer	Utilizes a Mixer backbone, which employs a permutation-equivariant model for processing data by mixing different channels and spatial dimensions.	[THK+21]
Transformer	A transformer-based classifier, utilizing multi-head attention mechanisms for capturing relationships across inputs, particularly useful in sequence-to-sequence tasks like text or time-series classification.	[VSP+17]
ExternalTransformer	A transformer classifier using an external transformer backbone with attention mechanisms designed for large-scale or external data processing.	[GLMH22]
ConvMixer	Uses a ConvMixer backbone, combining convolutional layers with mixers for efficiently processing high-dimensional input data, typically for image-based classification tasks.	[TK22]
AFTFull	Uses an AFTFull (Attention-Free Transformer) backbone, focusing on a full attention mechanism without requiring position bias, used in tasks requiring efficient large-scale data processing.	[ZTS+21]
ResidualAttention	A classifier with a residual attention backbone that stacks attention modules with residual connections, improving performance for very deep models and providing better object recognition.	[ZW21]
SimAM	A classifier using the SimAM (Simple Attention Mechanism) backbone, focusing on a simple yet effective attention mechanism with minimal computational overhead.	[LZLX21]

Model Name	Model Summary	Reference
SEAttention	A classifier utilizing the SEAttention (Squeeze-and-Excitation) mechanism, which adaptively recalibrates channel-wise feature responses to improve performance in image classification tasks.	[HSS18]
DoubleAttention	A classifier with a double attention backbone that aggregates and propagates informative features across both spatial and temporal dimensions, improving efficiency and performance.	[CKL+18]
PerformerAttention	Uses a Performer backbone with linear time and space complexity, leveraging a novel attention mechanism (FAVOR+) to approximate softmax attention, providing scalability in large tasks.	[CLD+20]
ParNetAttention	A classifier using the ParNet (Position-Aware Relation Network) backbone, which focuses on improving image-text matching by modeling both semantic and spatial relationships.	[GBDK22]
UFOAttention	A classifier using the UFO (Universal Feature-Oriented Attention) backbone, which focuses on attention mechanisms with flexible heads and dropout regularization to enhance performance.	[GBDK22]
ECAAttention	Uses an ECA (Efficient Channel Attention) backbone, providing efficient channel attention via a 1D convolution mechanism to improve performance without significant model complexity.	[WWZ+20]
CBAM	A classifier using the CBAM (Convolutional Block Attention Module) backbone, which combines channel and spatial attention to improve the quality of feature extraction in image classification tasks.	[WPLK18]
SwitchTransformer	A classifier using a Switch Transformer backbone, which incorporates a mixture of experts approach for more efficient computation, using only a subset of model parameters at each step.	[FZS22]

Table 1: Overview of models, their summaries, and references.

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